

Data Cleaning & Preprocessing Guide

How the Ingestion Engine Validates and Transforms Raw Files

1. Semantic Column Mapping

Users often upload spreadsheets with inconsistent headers (e.g. 'qty', 'units', 'counts').

CIVORA uses a semantic dictionary to automatically map files:

- * Product: 'item', 'name', 'product', 'goods', 'article'
- * Revenue: 'revenue', 'price', 'sales', 'earnings', 'income', 'total money'
- * Cost: 'cost', 'expense', 'expenses', 'outlay', 'spending'
- * Quantity: 'quantity', 'qty', 'count', 'amount', 'volume', 'units'
- * Region: 'region', 'area', 'location', 'state', 'city', 'place'

2. Regional State Normalization

Spelling discrepancies can break map visualizations. CIVORA maps region names to canonical Indian states using 'state_aliases.json':

- * E.g., 'Tamilnadu', 'TN', 'tamil nadu' -> 'Tamil Nadu'
- * E.g., 'mumbai', 'bombay', 'Maharashtra' -> 'Maharashtra'

If no match is found, the value is title-cased.

3. Data Type Casting and Safety

- * Numeric conversion: Revenue, cost, and quantity columns are parsed using `pandas.to_numeric()`. Non-numeric entries (e.g., 'NA', '-', '\$') are converted to 0.
- * Missing categories: Null categories default to 'General'.
- * Missing products: Null product names are handled, defaulting to 'N/A' or skipped.